



Minutes of Meeting of Programme Assessment Committee of the Department of Civil Engineering

Prof. (Dr.) Atul Kant Piyoosh welcomed all the respected members in the meeting.

The meeting of the Programme Assessment Committee of the department of Civil Engineering was held on 24 February 2024. The following are the PAC members:

1. Dr. Atul Kant Piyoosh	Convener
2. Er. Akhil Pandey	Member Secretary
3. Er. Neha Singh	Member
4. Er. Arjit Tiwari	Member
5. Er. Avneesh Tiwari	Member
6. Er. Akhilesh Rathore	Special Invitee (Internal)
7. Er. Naveen Kumar	Special Invitee (External)
8. Er. Shubham Srivastava	Alumni
9. Er. Prabhat Kumar Srivastava	Industrial Expert
10. Dr. Kumar Venkatesh	External Member

The following Agenda were discussed:

- Agenda-1:** Review, evaluate, and improve the alignment of the current Civil Engineering curriculum with present industrial requirements.
- Agenda-2:** Verification and approval of course outcomes (COs) and Course Outcome - Program Outcome/ Program Specific Outcome (CO-PO/PSO) mapping for subjects offered in B.Tech. Civil Engineering.
- Agenda-3:** Explore opportunities for interdisciplinary collaboration and integration within the B.Tech. Civil Engineering Curriculum to foster a holistic learning experience
- Agenda-4:** Facilitating publication opportunities for Final Year B. Tech Civil Engineering projects.
- Agenda-5:** Streamlining the process for finalizing topics of Webinars and Seminars.
- Agenda-6:** Strengthen laboratory practices to integrate modern techniques and technologies, aligning with industry standards and current Civil Engineering practices.
- Agenda-7:** Promote industry engagement by inculcating innovative and entrepreneurial skills.

Discussion:

Agenda-1: Review, evaluate, and improve the alignment of the current Civil Engineering curriculum with present industrial requirements.

During the discussion, several external members emphasized their limited familiarity with the current course curriculum. Consequently, it was decided that the Course Curriculum of civil engineering would be shared among all the members to facilitate extrapolation and assessment of its alignment with present industrial requirements. Additionally, Er. Prabhat Kumar Srivastava emphasized that your institute is a lifetime institutional member of the Indian Concrete Institute. It will provide valuable insights and resources to contribute towards improving the alignment of the curriculum with industrial needs.

Agenda-2: Verification and approval of course outcomes (COs) and Course Outcome - Program Outcome/ Program Specific Outcome (CO-PO/PSO) mapping for subjects offered in B. Tech. Civil Engineering.

Er. Shubham Kumar suggested that exposure of students to industry should be in association or link such that it serves the course outcomes through real-world problems. Exposure to field should be a mandatory outcome. Er. Akhilesh Rathore criticized the illustration of COs on presentation and suggested that Faculty made CO and Existed COs should be presented on side by side to analyze effectively. Newly Formulated /Modified Cos would be shared among all the external members for approval.

Agenda-3: Explore opportunities for interdisciplinary collaboration and integration within the B. Tech. Civil Engineering Curriculum to foster a holistic learning experience.

"Exploration within and beyond," as stated by Dr. Kumar Venkatesh, suggests fostering multidisciplinary collaboration among teams from various branches of the B.Tech program. For instance, the Computer Science department could collaborate with Civil Engineering on topics such as AI, IoT in civil engineering, and Neural Networks. Collaborating with the Mechanical Engineering department could involve working on topics like 3D printing and machine foundation. Research on Bioconcrete could involve collaboration with the Department of Biotechnology. Structural Health Monitoring could benefit from collaboration with the Electrical Engineering department. Dr. Venkatesh emphasized that interdisciplinary project teams should include members from various departments, not limited to Civil Engineering alone.

Moreover, interdisciplinary teams should collaborate within departments from different specializations. For instance, any project related to a Sewage Treatment Plant would necessitate the expertise of Environmental engineers, Geotechnical engineers, and Structural engineers working together.

Agenda-4: Facilitating publication opportunities for Final Year B. Tech Civil Engineering projects.

Er. Naveen Kumar provided valuable insights regarding opportunities for facilitating the publication of Final Year B. Tech Civil Engineering projects. He mentioned organizations such as CRRI, CDRI, IITs, CBRI Roorkee, and NCC, which host poster/presentation competitions. These platforms can serve as excellent avenues for students to showcase their projects and potentially publish their work.

Agenda-5: Streamlining the process for finalizing topics of Webinars and Seminars

Er. Naveen Kumar proposed that topics for Webinars and Seminars should be futuristic and should be organized by professionals and industry experts to ensure relevance and depth. Additionally, Er. Prabhat Kumar Srivastava emphasized that titles should align with sustainability goals, ensuring that the topics contribute to sustainable development. Er. Shubham Srivastava further suggested that students should be assigned assignments for every seminar or guest lecture to evaluate their learning abilities effectively. Moreover, workshops should be prioritized to provide hands-on learning experiences to the students.

Agenda-6: Strengthen laboratory practices to integrate modern techniques and technologies, aligning with industry standards and current Civil Engineering practices.

Er. Naveen Kumar emphasized the need to replace obsolete machinery in the laboratory with equipment and machinery conducive to research work. Additionally, Er. Prabhat Kumar Srivastava proposed the incorporation of Structural Health Monitoring (SHM) devices, Non-Destructive Testing (NDT) devices, and drones into the laboratory setup. These modern technologies would enhance laboratory practices and ensure alignment with industry standards and current Civil Engineering practices.

Agenda 7: Promote industry engagement by inculcating innovative and entrepreneurial skills.

Er. Naveen Kumar emphasized the importance of imparting skills to students that lead to creative and innovative outcomes. He proposed aligning research groups in the final year with industrial experts to provide industrial mentorship, fostering industry engagement. Dr. Kumar Venkatesh suggested the initiation of an incubation center, Industry-Institute Collaboration (IIC) cell, Research and Development Center (RDC), and Skill Development Programs (SDP). It was noted that these cells are already established within the department. Furthermore, Dr. Venkatesh recommended that the department should encourage the generation of "sellable ideas" from students, further promoting innovative and entrepreneurial skills among them.

Attendance Record

S.No.	Members	Designation	Attendance
1	Dr. Atul Kant Piyoosh	Convener	P
2	Er. Akhil Pandey	Member Secretary	P
3	Er. Neha Singh	Member	P
4	Er. Arjit Tiwari	Member	P
5	Er. Avneesh Tiwari	Member	P
6	Er. Akhilesh Rathore	Special Invitee (Internal)	P
7	Er. Naveen Kumar	Special Invitee (External)	P
8	Er. Shubham Srivastava	Alumni	P
9	Er. Prabhat Kumar Srivastava	Industrial Expert	P
10	Dr. Kumar Venkatesh	External Member	P


Signature of HoD:
24.02.24